Rahul Moka

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EDUCATION

University of Memphis Memphis, TN
Bachelor of Science in Computer Science; Dean's List *Aug. 2023 – Dec. 2028*

- **Relevant Coursework:** Intro to Artificial Intelligence, Design & Analysis of Algorithms, Computer Organization & Architecture, Discrete Structures

EXPERIENCE

Triad Tech May 2026 – Jul. 2026
Software Engineer Intern, Full Stack

- Developed an **MCP server** integrating **Slack and Linear** that automated the team's project-management and client-communication workflows, improving efficiency by an estimated **30%**
- Co-designed the **MVC architecture** and **SQL schema** for Cork-Board, a **Rails + React** project-management platform, building the **REST APIs** that centralized **7 projects across 4 clients**
- Reviewed and validated **AI-generated pull requests** under a two-reviewer merge policy

Freelance Jul. 2026
AI Engineer, Private RAG Infrastructure

- Engineered **RAG pipelines** for **finance and marketing clients**, unifying OCR'd scans, web and internal data into one **VPC-hosted PostgreSQL** corpus with hybrid **HNSW** (**pgvector**) and **BM25** retrieval
- Implemented custom **MCP servers** exposing retrieval and domain tools to **LLM agents**, so **numeric outputs were computed in code** rather than generated by the model
- Shipped a **FastAPI + React** GenAI app with streaming LLM answers, inline source citations and **role-scoped access**, deployed to **client-owned private cloud** so no proprietary data left their perimeter

PROJECTS

Transaction NLP: Merchant Matching & Categorization | *PyTorch, Hugging Face, FastAPI* Oct. 2026 – Nov. 2026

- Built a **data pipeline** that cleans raw descriptors and engineers features, resolving merchants via **RapidFuzz** at **95.9% top-1** on corrupted text vs. **57.8%** for sentence embeddings
- Fine-tuned **DistilBERT** (**Hugging Face**, **PyTorch**) to categorize transactions, reaching **45.3% ± 3.1%** accuracy on unseen merchants across 3 seeds vs. **17.6%** for a TF-IDF + logistic regression baseline
- Served predictions via **FastAPI** with human-review routing for low-confidence results, a model card, **27 pytest** tests and **GitHub Actions** CI

Personal Adaptive Scheduler | *Python, FastAPI, Claude API, TypeScript* Aug. 2026

- Built a **FastAPI + SQLite** backend that generates a fixed daily schedule and runs a **Claude-powered agent** twice a day to fold new coursework into it
- Built an **Expo (React Native + TypeScript)** app with a day timeline, calendar view and push notifications, deployed to a VPS with systemd timers and Caddy

Agent Evaluation & Benchmarking Harness | *Python, Claude API, Prompt Engineering* Oct. 2026

- Cut **cost per task by 65%** and **tokens by 72%** while raising accuracy from **86.8% to 92.5%**, by restructuring agent instructions and loading context only when needed
- Designed a test harness with **60 labeled test cases** comparing agent variants on accuracy (**95% confidence intervals**), cost and latency, with a **CI/CD** gate that blocks regressions

Linc: AI Life Insurance Assistant | *React, Next.js, OpenAI API* Oct. 2026

- Built with a **4-person team** at Lincoln Financial's CodeLinc 11 hackathon: an AI assistant that turns a conversation into a life-insurance coverage estimate, deployed on **AWS**
- Developed a **React** interface with **OpenAI API** chat and **ElevenLabs** voice that visualizes the coverage math and runs "what if" scenarios

TECHNICAL SKILLS

AI & GenAI: OpenAI API, Claude API, Prompt Engineering, RAG, MCP, LangGraph, LLM Evaluation, Amazon Bedrock; AI-assisted development with **Claude Code**, **GitHub Copilot**
ML & Data: PyTorch, Hugging Face, scikit-learn, LightGBM, Pandas, Feature Engineering, NLP, PostgreSQL / pgvector
Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Ruby, HTML, CSS
Frameworks & Cloud: React, Next.js, Node.js, FastAPI, Flask, Rails, REST APIs, AWS (Lambda, S3, DynamoDB), Docker, GitHub Actions